

scenario_A-C

August 26, 2026

Utilize XOpt to perform optimization on scenario A again.

```
[1]: import os, sys
current_dir = os.getcwd()
sys.path.append(os.path.join(current_dir, '..'))
# import importlib
# import experiments_utils
# importlib.reload(experiments_utils)
from experiments_utils import *
from configs import *
```

Scenario A with conventional optimization methods

```
[ ]: # # scenario A setting
# CURRENT_BOUNDS = (0.01, 1.5)
# EPSILON = 1e-3

# # parameter setting
# # N_RUNS_A = 1000
# N_RUNS_A = 1
# A_BEAMLINE_LEN = 10

# METHODS_A = [
#     ("Nelder-Mead", None),
#     ("L-BFGS-B", "2-point"),
#     ("SLSQP", "2-point"),
#     ("COBYLA", None),
#     ("trust-constr", "2-point"),
# ]

# A_VARS = {
#     1: ["I", "current", lambda x: x],
#     3: ["I2", "current", lambda x: x],
# }
# A_OBJ = {
#     8: [{"measure": ["x", "alpha"], "goal": 0.0, "weight": 1.0}],
```

```
# 9: [{"measure": ["y", "alpha"], "goal": 0.0, "weight": 1.0}],
# }
```

```
[ ]: # # default parameter version
# METHOD_OPTIONS = {
#     "L-BFGS-B": {
#         "maxcor": 10,
#         "ftol": 2.220446049250313e-09,
#         "gtol": 1e-5,
#         "eps": 1e-8,
#         "maxfun": 15000,
#         "maxiter": 15000,
#         "maxls": 20,
#     },
#     "SLSQP": {
#         "maxiter": 100,
#         "ftol": 1e-6,
#         "iprint": 1,
#         "disp": False,
#         "eps": 1.4901161193847656e-08,
#         "finite_diff_rel_step": None,
#         "workers": None,
#     },
#     "trust-constr": {
#         "xtol": 1e-8,
#         "gtol": 1e-8,
#         "barrier_tol": 1e-8,
#         "sparse_jacobian": None,
#         "maxiter": 1000,
#         "verbose": 0,
#         "finite_diff_rel_step": None,
#         "initial_constr_penalty": 1.0,
#         "initial_tr_radius": 1.0,
#         "initial_barrier_parameter": 0.1,
#         "initial_barrier_tolerance": 0.1,
#         "factorization_method": None,
#         "disp": False,
#         "workers": None,
#     },
#     "Nelder-Mead": {
#         "maxiter": None,
#         "maxfev": None,
#         "disp": False,
#         "return_all": False,
#         "initial_simplex": None,
#         "xatol": 1e-4,
#         "fatol": 1e-4,
```

```
#         "adaptive": False,
#     },
#     "COBYLA": {
#         "rhobeg": 1.0,
#         "tol": 1e-4,
#         "maxiter": 1000,
#         "disp": 0,
#         "catol": None,
#         "f_target": float("-inf"),
#     },
# }
```

```
[4]: results_A, df_A, final_summary = run_scenario_A(
    CURRENT_BOUNDS,
    EPSILON,
    N_RUNS_A,
    FELSIM_S1_CURRENTS,
    A_OBJ,
    A_VARS,
    A_BEAMLINE_LEN,
    METHODS_A,
    scale="log",
    METHOD_OPTIONS=METHOD_OPTIONS,
)
display_results(results_A, A_VARS, FELSIM_S1_CURRENTS)
```

```
use_log: False
use_noise: False
```

```
Scenario A - Nelder-Mead (1 runs)
Nelder-Mead          run 1/1  MSE=3.900e-08  nit=33  nfev=63  t=13.06s
```

```
Scenario A - L-BFGS-B+jac (1 runs)
L-BFGS-B             run 1/1  MSE=6.775e-14  nit=9   nfev=48  t=8.02s
```

```
Scenario A - SLSQP+jac (1 runs)
SLSQP                run 1/1  MSE=2.198e-07  nit=8   nfev=26  t=3.98s
```

```
Scenario A - COBYLA (1 runs)
COBYLA               run 1/1  MSE=2.484e-05  nit=-1  nfev=199  t=38.68s
```

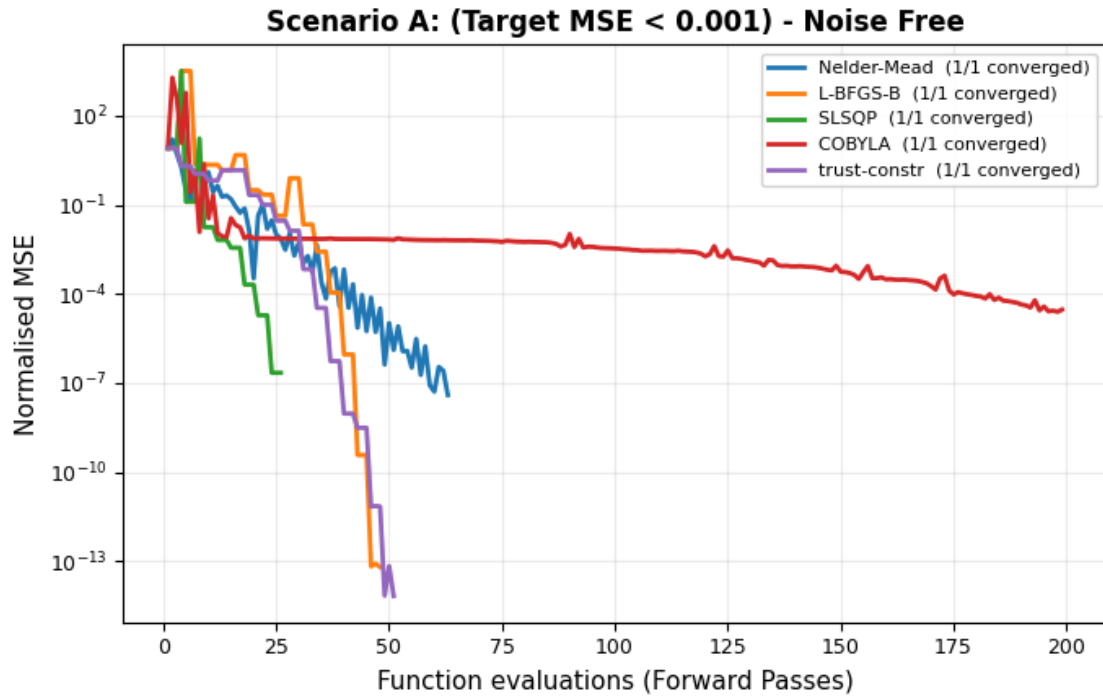
```
Scenario A - trust-constr+jac (1 runs)
trust-constr         run 1/1  MSE=6.951e-15  nit=22  nfev=51  t=9.92s
```

```
=====
SCENARIO A: Robustness (Tol < 0.001) & Discovered Physics State vs Reference
=====
```

```

=====
method_tag  conv_rate nit_mean nfev_mean wall_time_mean final_mse_mean    I
I_ref      I2      I2_ref  8_alpha_x 9_alpha_y
COBYLA      100%    199.0    199.0    38.684 s      2.48e-05    0.8371 A
0.8218 A 1.1217 A 1.0430 A   0.0061   -0.0046
L-BFGS-B    100%     9.0     48.0     8.018 s      6.77e-14    0.7556 A
0.8218 A 0.5852 A 1.0430 A   0.0000   -0.0000
Nelder-Mead 100%    33.0    63.0    13.062 s      3.90e-08    0.8352 A
0.8218 A 1.1205 A 1.0430 A   0.0003   -0.0001
SLSQP       100%     8.0    26.0     3.981 s      2.20e-07    0.8349 A
0.8218 A 1.1203 A 1.0430 A  -0.0006    0.0002
trust-constr 100%    22.0    51.0     9.922 s      6.95e-15    0.7556 A
0.8218 A 0.5852 A 1.0430 A  -0.0000   -0.0000
=====
=====

```



Scenario A: Best Optimised Quadrupole Currents per Method

Method	Best MSE	I (A)	I2 (A)
Nelder-Mead	3.9002e-08	0.8352	1.1205
L-BFGS-B	6.7748e-14	0.7556	0.5852
SLSQP	2.1980e-07	0.8349	1.1203
COBYLA	2.4836e-05	0.8371	1.1217
trust-constr	6.9511e-15	0.7556	0.5852

Reference FELsim S1 currents: I=0.8218 A, I2=1.0430 A

Overall best across all methods/runs:

Method: trust-constr, MSE=6.9511e-15

I=0.7556 A I2=0.5852 A

```
[4]: {'I': np.float64(0.7555847866150749), 'I2': np.float64(0.5852131177800047)}
```

```
[5]: # save the results
# experiment_save_path = os.path.join(CURRENT_DIR, "../results/
# ↪scenario_A_results_v4.pkl")
# save_results(results_A, df_A, final_summary, experiment_save_path)
```

```
[6]: # load the results
# experiment_save_path = os.path.join(CURRENT_DIR, "../results/
# ↪scenario_A_results_v4.pkl")
# results_A, df_A, final_summary = load_results(experiment_save_path)
```

scenario B with conventional optimization methods

```
[ ]: # # scenario B setting
# # CURRENT_BOUNDS_BC = (0.01, 5.0)
# # EPSILON = 1e-3

# # REF_I_1 = 0.8218
# # REF_I_3 = 1.0430
# # REF_I = [REF_I_1, REF_I_3]

# # parameter setting
# # N_RUNS_B = 1000
# N_RUNS_B = 1
# A_BEAMLINE_LEN = 138

# METHODS_B = [
#     ("Nelder-Mead", None),
#     ("L-BFGS-B", "2-point"),
#     ("SLSQP", "2-point"),
#     ("COBYLA", None),
#     ("trust-constr", "2-point"),
# ]

# # variables and constraints for scenario B
# def _v(name):
#     return [name, "current", lambda x, _n=name: x]

# STAGES_B = [
#     (
#         "Stage 1 Doublet",
```

```

#         {1: _v("I"), 3: _v("I2")},
#         {
#             8: [
#                 {"measure": ["x", "alpha"], "goal": 0, "weight": 1},
#                 {"measure": ["x", "beta"], "goal": 0.1, "weight": 0.0},
#             ],
#             9: [
#                 {"measure": ["y", "alpha"], "goal": 0, "weight": 1},
#                 {"measure": ["y", "beta"], "goal": 0.1, "weight": 0.5},
#             ],
#         },
#         10,
#         {
#             "I": {"bounds": CURRENT_BOUNDS_BC, "start": 1},
#             "I2": {"bounds": CURRENT_BOUNDS_BC, "start": 1},
#         },
#     ),
#     (
#         "Stage 2 Chrom.1",
#         {10: _v("I")},
#         {15: [{"measure": ["x", "dispersion"], "goal": 0, "weight": 1}]},
#         16,
#         {"I": {"bounds": CURRENT_BOUNDS_BC, "start": 1}},
#     ),
#     (
#         "Stage 3 Triplet1",
#         {16: _v("I"), 18: _v("I2"), 20: _v("I3")},
#         {
#             25: [
#                 {"measure": ["x", "alpha"], "goal": 0, "weight": 1},
#                 {"measure": ["x", "beta"], "goal": 0.1, "weight": 0.5},
#             ],
#             26: [
#                 {"measure": ["y", "alpha"], "goal": 0, "weight": 1},
#                 {"measure": ["y", "beta"], "goal": 0.1, "weight": 0.5},
#             ],
#         },
#         27,
#         {
#             "I": {"bounds": CURRENT_BOUNDS_BC, "start": 2},
#             "I2": {"bounds": CURRENT_BOUNDS_BC, "start": 5},
#             "I3": {"bounds": CURRENT_BOUNDS_BC, "start": 3},
#         },
#     ),
#     (
#         "Stage 4 Chrom.2",
#         {27: _v("I")},

```

```

#         {32: [{"measure": ["x", "dispersion"], "goal": 0, "weight": 1}]},
#         33,
#         {"I": {"bounds": CURRENT_BOUNDS_BC, "start": 1}},
#     ),
#     (
#         "Stage 5 DblTriplet",
#         {37: _v("I"), 35: _v("I2"), 33: _v("I3")},
#         {
#             37: [
#                 {"measure": ["x", "alpha"], "goal": 0, "weight": 1},
#                 {"measure": ["y", "alpha"], "goal": 0, "weight": 1},
#                 {"measure": ["x", "envelope"], "goal": 2.0, "weight": 1},
#                 {"measure": ["y", "envelope"], "goal": 2.0, "weight": 1},
#             ]
#         },
#         38,
#         {
#             "I": {"bounds": CURRENT_BOUNDS_BC, "start": 0.28},
#             "I2": {"bounds": CURRENT_BOUNDS_BC, "start": 2.65},
#             "I3": {"bounds": CURRENT_BOUNDS_BC, "start": 2.69},
#         },
#     ),
#     (
#         "Stage 6 Chrom.3",
#         {50: _v("I")},
#         {55: [{"measure": ["x", "dispersion"], "goal": 0, "weight": 1}]},
#         56,
#         {"I": {"bounds": CURRENT_BOUNDS_BC, "start": 1}},
#     ),
#     (
#         "Stage 7 IP",
#         {56: _v("I"), 58: _v("I2")},
#         {
#             59: [
#                 {"measure": ["x", "envelope"], "goal": 0.0, "weight": 1},
#                 {"measure": ["y", "envelope"], "goal": 0.0, "weight": 1},
#             ]
#         },
#         60,
#         {
#             "I": {"bounds": CURRENT_BOUNDS_BC, "start": 2},
#             "I2": {"bounds": CURRENT_BOUNDS_BC, "start": 2},
#         },
#     ),
#     (
#         "Stage 8 Doublet2",
#         {61: _v("I"), 63: _v("I2")},

```

```

#      {
#          68: [
#              {"measure": ["x", "alpha"], "goal": 0, "weight": 1},
#              {"measure": ["x", "beta"], "goal": 0.1, "weight": 0.5},
#          ],
#          69: [
#              {"measure": ["y", "alpha"], "goal": 0, "weight": 1},
#              {"measure": ["y", "beta"], "goal": 0.1, "weight": 0.5},
#          ],
#      },
#      70,
#      {
#          "I": {"bounds": CURRENT_BOUNDS_BC, "start": 2},
#          "I2": {"bounds": CURRENT_BOUNDS_BC, "start": 2},
#      },
#  ),
#  (
#      "Stage 9 Chrom.4",
#      {70: _v("I")},
#      {75: [{"measure": ["x", "dispersion"], "goal": 0, "weight": 1}]},
#      76,
#      {"I": {"bounds": CURRENT_BOUNDS_BC, "start": 1}},
#  ),
#  (
#      "Stage 10 Triplet3",
#      {76: _v("I"), 78: _v("I2"), 80: _v("I3")},
#      {
#          85: [
#              {"measure": ["x", "alpha"], "goal": 0, "weight": 1},
#              {"measure": ["x", "beta"], "goal": 0.1, "weight": 0.5},
#          ],
#          86: [
#              {"measure": ["y", "alpha"], "goal": 0, "weight": 1},
#              {"measure": ["y", "beta"], "goal": 0.1, "weight": 0.5},
#          ],
#      },
#      87,
#      {
#          "I": {"bounds": CURRENT_BOUNDS_BC, "start": 2},
#          "I2": {"bounds": CURRENT_BOUNDS_BC, "start": 2},
#          "I3": {"bounds": CURRENT_BOUNDS_BC, "start": 2},
#      },
#  ),
#  (
#      "Stage 11 UND Match",
#      {87: _v("Ic"), 93: _v("I"), 95: _v("I2"), 97: _v("I3")},
#      {

```



```

#          92: [{"measure": ["x", "dispersion"], "goal": 0, "weight": 0.5}],
#          117: [
#              {"measure": ["x", "alpha"], "goal": ALPHA_XM, "weight": 1},
#              {"measure": ["y", "alpha"], "goal": ALPHA_YM, "weight": 1},
#              {"measure": ["x", "beta"], "goal": BETA_XM, "weight": 1},
#              {"measure": ["y", "beta"], "goal": BETA_YM, "weight": 1},
#          ],
#      },
#      118,
#      {
#          "Ic": {"bounds": CURRENT_BOUNDS_BC, "start": 4},
#          "I": {"bounds": CURRENT_BOUNDS_BC, "start": 2},
#          "I2": {"bounds": CURRENT_BOUNDS_BC, "start": 2},
#          "I3": {"bounds": CURRENT_BOUNDS_BC, "start": 2},
#      },
#  ),
# ]

```

```
[8]: from scenario_B_utils import run_scenario_B
```

```

results_B, df_B, final_beamline_B = run_scenario_B(
    STAGES_B=STAGES_B,
    PARTICLES=PARTICLES,
    EXCEL_PATH=EXCEL_PATH,
    METHODS_B=METHODS_B,
    N_RUNS_B=N_RUNS_B,
    METHOD_OPTIONS=METHOD_OPTIONS,
    verbose=True
)

display(df_B)

```

Starting Scenario B (11-Stage Sequential) - Method: Nelder-Mead (1 runs total)

Stage 1 Doublet	MSE=1.708e-04	nfev=61	t=11.69s
Stage 2 Chrom.1	MSE=1.653e-14	nfev=40	t=17.79s
Stage 3 Triplet1	MSE=1.255e-03	nfev=124	t=100.99s
Stage 4 Chrom.2	MSE=9.821e-06	nfev=16	t=15.67s
Stage 5 Db1Triplet	MSE=8.344e-02	nfev=443	t=489.56s
Stage 6 Chrom.3	MSE=4.003e-06	nfev=16	t=25.83s
Stage 7 IP	MSE=1.406e-02	nfev=88	t=142.74s
Stage 8 Doublet2	MSE=5.362e+01	nfev=80	t=161.59s
Stage 9 Chrom.4	MSE=5.965e-05	nfev=16	t=33.04s
Stage 10 Triplet3	MSE=1.480e-03	nfev=179	t=399.24s
Stage 11 UND Match	MSE=6.839e-01	nfev=800	t=2897.78s

Nelder-Mead pipeline execution completed, total time: 4296.11s

Starting Scenario B (11-Stage Sequential) - Method: L-BFGS-B (1 runs total)

Stage 1 Doublet	MSE=1.306e-04	nfev=51	t=10.07s
Stage 2 Chrom.1	MSE=1.436e-09	nfev=10	t=4.66s
Stage 3 Triplet1	MSE=1.218e-03	nfev=56	t=40.85s
Stage 4 Chrom.2	MSE=1.005e-05	nfev=8	t=8.95s
Stage 5 Db1Triplet	MSE=1.585e-01	nfev=328	t=310.94s
Stage 6 Chrom.3	MSE=1.966e-08	nfev=10	t=14.90s
Stage 7 IP	MSE=1.401e-02	nfev=36	t=56.91s
Stage 8 Doublet2	MSE=2.031e+01	nfev=126	t=211.75s
Stage 9 Chrom.4	MSE=7.880e-05	nfev=8	t=15.34s
Stage 10 Triplet3	MSE=2.259e-03	nfev=152	t=340.35s
Stage 11 UND Match	MSE=6.643e+01	nfev=150	t=442.13s

L-BFGS-B pipeline execution completed, total time: 1457.04s

Starting Scenario B (11-Stage Sequential) - Method: SLSQP (1 runs total)

Stage 1 Doublet	MSE=1.307e-04	nfev=42	t=6.68s
Stage 2 Chrom.1	MSE=6.144e-04	nfev=2	t=0.72s
Stage 3 Triplet1	MSE=2.638e-03	nfev=99	t=59.57s
Stage 4 Chrom.2	MSE=1.545e-04	nfev=14	t=12.41s
Stage 5 Db1Triplet	MSE=5.688e-01	nfev=193	t=163.76s
Stage 6 Chrom.3	MSE=1.748e-09	nfev=16	t=21.23s
Stage 7 IP	MSE=1.999e-02	nfev=27	t=38.03s
Stage 8 Doublet2	MSE=6.962e-01	nfev=26	t=42.37s
Stage 9 Chrom.4	MSE=4.525e-04	nfev=2	t=3.60s
Stage 10 Triplet3	MSE=2.305e-03	nfev=126	t=259.97s
Stage 11 UND Match	MSE=8.701e-04	nfev=181	t=492.29s

SLSQP pipeline execution completed, total time: 1100.79s

Starting Scenario B (11-Stage Sequential) - Method: COBYLA (1 runs total)

Stage 1 Doublet	MSE=2.325e-04	nfev=30	t=4.59s
Stage 2 Chrom.1	MSE=3.433e-16	nfev=29	t=10.30s
Stage 3 Triplet1	MSE=2.111e-03	nfev=75	t=44.28s
Stage 4 Chrom.2	MSE=1.013e-05	nfev=7	t=6.11s
Stage 5 Db1Triplet	MSE=1.110e-01	nfev=194	t=174.41s
Stage 6 Chrom.3	MSE=1.333e-14	nfev=29	t=43.73s
Stage 7 IP	MSE=9.613e-04	nfev=129	t=195.89s
Stage 8 Doublet2	MSE=2.588e+01	nfev=56	t=96.63s
Stage 9 Chrom.4	MSE=2.411e-04	nfev=6	t=13.06s
Stage 10 Triplet3	MSE=2.326e-03	nfev=337	t=800.12s
Stage 11 UND Match	MSE=2.165e-02	nfev=242	t=1031.35s

COBYLA pipeline execution completed, total time: 2420.59s

Starting Scenario B (11-Stage Sequential) - Method: trust-constr (1 runs total)

Stage 1 Doublet	MSE=1.306e-04	nfev=153	t=51.26s
Stage 2 Chrom.1	MSE=5.265e-11	nfev=16	t=15.01s

Stage 3 Triplet1	MSE=1.218e-03	nfev=700	t=626.73s
Stage 4 Chrom.2	MSE=1.062e-05	nfev=18	t=28.32s
Stage 5 DblTriplet	MSE=1.596e-01	nfev=868	t=1272.05s
Stage 6 Chrom.3	MSE=1.365e-10	nfev=22	t=40.63s
Stage 7 IP	MSE=1.401e-02	nfev=36	t=84.25s
Stage 8 Doublet2	MSE=2.052e+01	nfev=399	t=1464.72s
Stage 9 Chrom.4	MSE=8.264e-05	nfev=14	t=44.01s
Stage 10 Triplet3	MSE=1.524e-03	nfev=2684	t=6537.59s
Stage 11 UND Match	MSE=5.116e-01	nfev=4270	t=11741.77s

trust-constr pipeline execution completed, total time: 21906.57s

	run	stage	stage_name	method_tag	final_mse	nfev	nit	\
0	1	1	Stage 1 Doublet	Nelder-Mead	1.707693e-04	61	33	
1	1	2	Stage 2 Chrom.1	Nelder-Mead	1.653135e-14	40	20	
2	1	3	Stage 3 Triplet1	Nelder-Mead	1.254534e-03	124	65	
3	1	4	Stage 4 Chrom.2	Nelder-Mead	9.820682e-06	16	8	
4	1	5	Stage 5 DblTriplet	Nelder-Mead	8.344140e-02	443	251	
5	1	6	Stage 6 Chrom.3	Nelder-Mead	4.003195e-06	16	8	
6	1	7	Stage 7 IP	Nelder-Mead	1.406229e-02	88	46	
7	1	8	Stage 8 Doublet2	Nelder-Mead	5.361675e+01	80	42	
8	1	9	Stage 9 Chrom.4	Nelder-Mead	5.964719e-05	16	8	
9	1	10	Stage 10 Triplet3	Nelder-Mead	1.480213e-03	179	97	
10	1	11	Stage 11 UND Match	Nelder-Mead	6.838857e-01	800	489	
11	1	1	Stage 1 Doublet	L-BFGS-B	1.306461e-04	51	12	
12	1	2	Stage 2 Chrom.1	L-BFGS-B	1.436269e-09	10	4	
13	1	3	Stage 3 Triplet1	L-BFGS-B	1.218138e-03	56	9	
14	1	4	Stage 4 Chrom.2	L-BFGS-B	1.004895e-05	8	3	
15	1	5	Stage 5 DblTriplet	L-BFGS-B	1.585234e-01	328	66	
16	1	6	Stage 6 Chrom.3	L-BFGS-B	1.966483e-08	10	4	
17	1	7	Stage 7 IP	L-BFGS-B	1.400762e-02	36	8	
18	1	8	Stage 8 Doublet2	L-BFGS-B	2.030763e+01	126	11	
19	1	9	Stage 9 Chrom.4	L-BFGS-B	7.879680e-05	8	3	
20	1	10	Stage 10 Triplet3	L-BFGS-B	2.258539e-03	152	27	
21	1	11	Stage 11 UND Match	L-BFGS-B	6.642922e+01	150	23	
22	1	1	Stage 1 Doublet	SLSQP	1.306833e-04	42	12	
23	1	2	Stage 2 Chrom.1	SLSQP	6.143946e-04	2	1	
24	1	3	Stage 3 Triplet1	SLSQP	2.638160e-03	99	24	
25	1	4	Stage 4 Chrom.2	SLSQP	1.545478e-04	14	7	
26	1	5	Stage 5 DblTriplet	SLSQP	5.687769e-01	193	44	
27	1	6	Stage 6 Chrom.3	SLSQP	1.747802e-09	16	8	
28	1	7	Stage 7 IP	SLSQP	1.999473e-02	27	9	
29	1	8	Stage 8 Doublet2	SLSQP	6.961861e-01	26	8	
30	1	9	Stage 9 Chrom.4	SLSQP	4.524884e-04	2	1	
31	1	10	Stage 10 Triplet3	SLSQP	2.304640e-03	126	30	
32	1	11	Stage 11 UND Match	SLSQP	8.700916e-04	181	34	
33	1	1	Stage 1 Doublet	COBYLA	2.324535e-04	30	30	
34	1	2	Stage 2 Chrom.1	COBYLA	3.432530e-16	29	29	
35	1	3	Stage 3 Triplet1	COBYLA	2.111397e-03	75	75	

36	1	4	Stage 4 Chrom.2	COBYLA	1.013433e-05	7	7
37	1	5	Stage 5 Db1Triplet	COBYLA	1.109767e-01	194	194
38	1	6	Stage 6 Chrom.3	COBYLA	1.332687e-14	29	29
39	1	7	Stage 7 IP	COBYLA	9.612973e-04	129	129
40	1	8	Stage 8 Doublet2	COBYLA	2.588399e+01	56	56
41	1	9	Stage 9 Chrom.4	COBYLA	2.410705e-04	6	6
42	1	10	Stage 10 Triplet3	COBYLA	2.325833e-03	337	337
43	1	11	Stage 11 UND Match	COBYLA	2.164503e-02	242	242
44	1	1	Stage 1 Doublet	trust-constr	1.306461e-04	153	57
45	1	2	Stage 2 Chrom.1	trust-constr	5.265401e-11	16	17
46	1	3	Stage 3 Triplet1	trust-constr	1.217720e-03	700	135
47	1	4	Stage 4 Chrom.2	trust-constr	1.061924e-05	18	17
48	1	5	Stage 5 Db1Triplet	trust-constr	1.596221e-01	868	201
49	1	6	Stage 6 Chrom.3	trust-constr	1.364918e-10	22	20
50	1	7	Stage 7 IP	trust-constr	1.401170e-02	36	18
51	1	8	Stage 8 Doublet2	trust-constr	2.051930e+01	399	123
52	1	9	Stage 9 Chrom.4	trust-constr	8.263531e-05	14	14
53	1	10	Stage 10 Triplet3	trust-constr	1.524190e-03	2684	593
54	1	11	Stage 11 UND Match	trust-constr	5.115784e-01	4270	839

	wall_time	success
0	11.690	True
1	17.790	True
2	100.989	True
3	15.667	True
4	489.562	True
5	25.831	True
6	142.740	True
7	161.589	True
8	33.038	True
9	399.235	True
10	2897.778	False
11	10.069	True
12	4.657	True
13	40.847	True
14	8.946	True
15	310.939	True
16	14.898	True
17	56.912	True
18	211.745	False
19	15.343	True
20	340.354	True
21	442.126	True
22	6.681	True
23	0.724	True
24	59.575	True
25	12.406	True
26	163.757	True

27	21.228	True
28	38.032	True
29	42.372	True
30	3.597	True
31	259.970	True
32	492.290	True
33	4.590	True
34	10.299	True
35	44.283	True
36	6.113	True
37	174.413	True
38	43.732	True
39	195.892	True
40	96.634	True
41	13.059	True
42	800.121	True
43	1031.352	True
44	51.256	True
45	15.007	True
46	626.734	True
47	28.319	True
48	1272.051	True
49	40.631	True
50	84.245	True
51	1464.721	True
52	44.012	True
53	6537.587	True
54	11741.771	True

scenario C with conventional optimization methods

```
[ ]: # # =====
# #   Scenario C Execution (Combined 11-parameter)
# #   =====
# print("Running Scenario C: Combined ~11-parameter (IP+UND)...")

# CURRENT_BOUNDS_BC = (0.01, 5.0) # Expanded bounds since we are dealing with a
↳larger optimization space
# EPSILON = 1e-3
# N_RUNS_C = 1
# C_BEAMLINE_LEN = 118

# METHODS_C = [
#     ("Nelder-Mead", None),
#     ("L-BFGS-B", "2-point"),
#     ("SLSQP", "2-point"),
#     ("COBYLA", None),
#     ("trust-constr", "2-point"),
```

```

# ]

# # Ensure variables are uniquely named to prevent DataFrame column collisions
# def _v_C(name): return [name, "current", lambda x, _n=name: x]

# C_VARS = {
#     56: _v_C("I_56"), 58: _v_C("I_58"),
#     61: _v_C("I_61"), 63: _v_C("I_63"),
#     76: _v_C("I_76"), 78: _v_C("I_78"), 80: _v_C("I_80"),
#     87: _v_C("I_87"), 93: _v_C("I_93"), 95: _v_C("I_95"), 97: _v_C("I_97")
# }

# # Objectives: IP spot + UND beta matching
# C_OBJ = {
#     59: [
#         {"measure": ["x", "envelope"], "goal": 0.0, "weight": 1.0},
#         {"measure": ["y", "envelope"], "goal": 0.0, "weight": 1.0},
#     ],
#     117: [
#         {"measure": ["x", "alpha"], "goal": ALPHA_XM, "weight": 1.0},
#         {"measure": ["y", "alpha"], "goal": ALPHA_YM, "weight": 1.0},
#         {"measure": ["x", "beta"], "goal": BETA_XM, "weight": 1.0},
#         {"measure": ["y", "beta"], "goal": BETA_YM, "weight": 1.0},
#     ]
# }

```

Running Scenario C: Combined ~11-parameter (IP+UND)...

use_log: False | use_noise: False

Scenario C - Nelder-Mead (1 runs)

Nelder-Mead run 1/1 MSE=7.222e+13 nit=290 nfev=937 t=2736.83s

Scenario C - L-BFGS-B+jac (1 runs)

L-BFGS-B run 1/1 MSE=3.386e+32 nit=1 nfev=408 t=1157.39s

Scenario C - SLSQP+jac (1 runs)

SLSQP run 1/1 MSE=3.388e+32 nit=1 nfev=12 t=34.18s

Scenario C - COBYLA (1 runs)

COBYLA run 1/1 MSE=1.216e+14 nit=-1 nfev=90 t=256.34s

Scenario C - trust-constr+jac (1 runs)

trust-constr run 1/1 MSE=1.033e+15 nit=183 nfev=3768 t=11264.16s

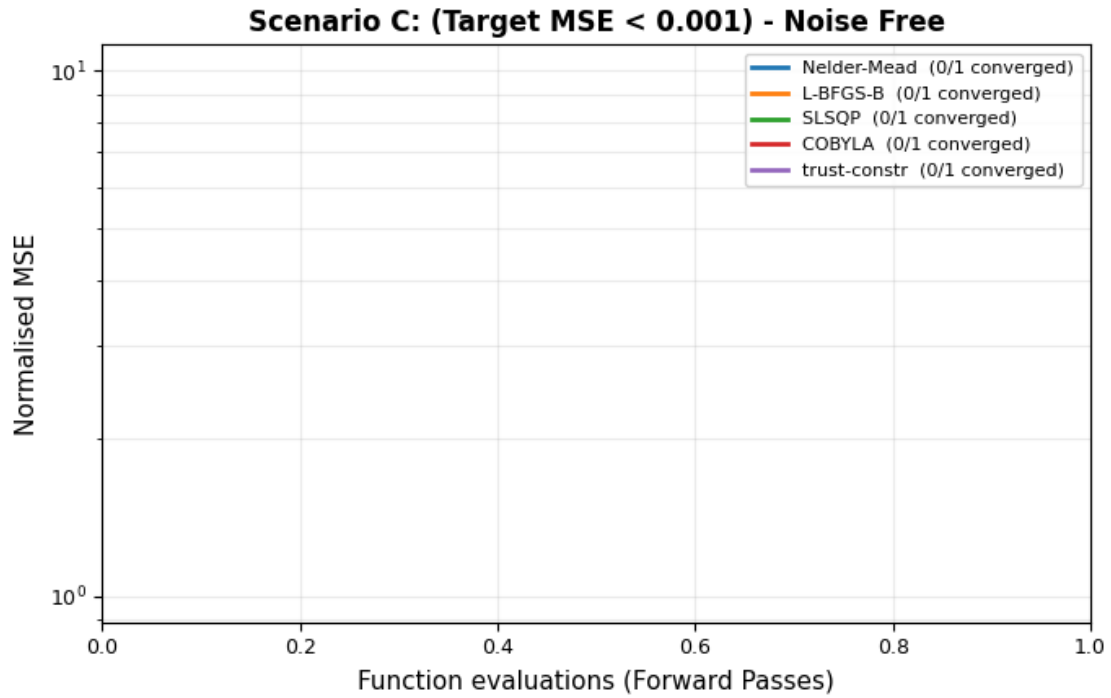
=====

SCENARIO C: Robustness (Tol < 0.001) & Discovered Physics State vs Reference

```

=====
=====
method_tag  conv_rate nit_mean nfev_mean wall_time_mean final_mse_mean  I_56
I_56_ref    I_58    I_58_ref    I_61    I_61_ref    I_63    I_63_ref    I_76    I_76_ref
I_78    I_78_ref    I_80    I_80_ref    I_87    I_87_ref    I_93    I_93_ref    I_95
I_95_ref    I_97    I_97_ref 59_envelope_x      117_alpha_x
      COBYLA      0%      90.0      90.0      256.336 s      NaN      3.8785 A
3.1219 A 2.2574 A 3.3129 A 0.5412 A 5.1775 A 3.9325 A 4.0434 A 3.7000 A 3.9336 A
3.3724 A 4.0787 A 0.3719 A 0.0139 A 4.2944 A 1.3624 A 3.3818 A 0.9452 A 1.7522 A
2.8851 A 4.8784 A 2.1921 A      10.5253      -19836951.6309
      L-BFGS-B      0%      1.0      408.0      1157.392 s      NaN      3.8720 A
3.1219 A 2.2575 A 3.3129 A 0.6493 A 5.1775 A 3.9324 A 4.0434 A 3.8080 A 3.9336 A
2.2000 A 4.0787 A 0.4799 A 0.0139 A 4.2944 A 1.3624 A 3.4898 A 0.9452 A 1.8603 A
2.8851 A 4.8784 A 2.1921 A      10.5072      -40650030.7312
      Nelder-Mead      0%      290.0      937.0      2736.831 s      NaN      3.8692 A
3.1219 A 2.2613 A 3.3129 A 0.6516 A 5.1775 A 3.9450 A 4.0434 A 3.9132 A 3.9336 A
2.1846 A 4.0787 A 0.4810 A 0.0139 A 4.2807 A 1.3624 A 3.4827 A 0.9452 A 1.9060 A
2.8851 A 4.8784 A 2.1921 A      10.4702      -10481432.4170
      SLSQP      0%      1.0      12.0      34.185 s      NaN      3.8720 A
3.1219 A 2.2574 A 3.3129 A 0.6493 A 5.1775 A 3.9325 A 4.0434 A 3.8081 A 3.9336 A
2.2000 A 4.0787 A 0.4799 A 0.0139 A 4.2944 A 1.3624 A 3.4899 A 0.9452 A 1.8603 A
2.8851 A 4.8784 A 2.1921 A      10.5076      -23003316499334688.0000
trust-constr      0%      183.0      3768.0      11264.157 s      NaN      3.9971 A
3.1219 A 2.1701 A 3.3129 A 0.7321 A 5.1775 A 3.6431 A 4.0434 A 3.8267 A 3.9336 A
2.0477 A 4.0787 A 0.5988 A 0.0139 A 4.4440 A 1.3624 A 3.5190 A 0.9452 A 2.1764 A
2.8851 A 4.6599 A 2.1921 A      11.5303      -24657034502882068.0000
=====
=====

```



```
[ ]: # Run it through the universal single-stage runner
results_C, df_C, final_summary_C = run_scenario_C(
    scenario_name="C",
    CURRENT_BOUNDS=CURRENT_BOUNDS_BC,
    EPSILON=EPSILON,
    N_RUNS=N_RUNS_C,
    REF_CURRENTS=FELSIM_S1_CURRENTS,
    OBJ=C_OBJ,
    VARS=C_VARS,
    BEAMLINE_LEN=C_BEAMLINE_LEN,
    METHODS=METHODS_C,
    METHOD_OPTIONS=METHOD_OPTIONS,
    verbose=True
)
```